Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 October 2026
Team ID	NA
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Use the EV market file ev_market_test.csv (100,000 records, 37 attributes such as category, manufacturer, region, battery type, range, CO_2 saved, sales and rating) and connect it to Tableau as an extract.	High
2	Data Processing	Check data types, nulls and duplicates in Tableau; set Manufacturing Country as a geographic field; create a Hyper extract.	High
3	Business Analysis	Aggregate annual sales (SUM) by category, region, manufacturer and model year; average range, CO_2 savings and customer rating (AVG) by category or manufacturer; count vehicles by battery type; Top-10 rankings.	High
4	Business Insights & Analytical Outputs	8 worksheets, a dashboard with 6 charts and a Tableau Story that answer the questions of the three scenarios.	High
5	User Interaction & Analysis	Market Region filter, sorted bars, tooltips with exact values, story navigation.	Medium
6	Data Update & Reporting	Dashboard and story published on Tableau Public; refreshing the extract updates every chart.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Clear chart titles, sorted bars and a simple 2 × 3 dashboard layout that anyone can read.
NFR-2	**Security**	Only public, non-personal market data; dashboard is read-only on Tableau Public.
NFR-3	**Reliability**	Data checked for duplicates (0) and nulls; values in the charts match the source data.
NFR-4	**Performance**	Extract of 100,000 rows loads all charts in a few seconds.
NFR-5	**Availability**	Available 24×7 through the Tableau Public link.
NFR-6	**Scalability**	New data can be added by refreshing the extract; layout supports more charts.